EGYPT TOURISM INTELLIGENCE

PROJECT PLANNING & MANAGEMENT DOCUMENTATION

1. Project Proposal

1.1 Project Overview

This project analyzes Egypt's tourism using available data on arrivals, revenue, source markets, seasonality, accommodation, and selected landmarks. It identifies trends and supports recommendations with evidence.

1.2 Problem Statement

Arrivals alone do not show tourism's full value. The project also examines spending, length of stay, seasonality, and destination performance, based on available and comparable data.

1.3 Objectives

- Measure changes in tourist arrivals, tourism revenue, and tourist nights over time.
- Analyze source markets and compare their contribution to visitor volume and available economic indicators.
- Identify seasonal patterns and periods of relatively high or low demand.
- Develop a dedicated analytical page for each selected landmark or destination.
- Compare visitor and economic indicators only where source definitions and reporting levels are compatible.
- Produce recommendations supported by the analytical findings.

1.4 Project Scope

Included in Scope	Excluded from Scope
Tourism data analysis; SQL data preparation; Power BI reporting; landmark profiles; evidence-based recommendations.	Booking platform; personal tourist data; unsupported landmark revenue estimates; unverified metrics.

1.5 Data Limitations

The analysis depends on available data quality and detail. National, regional, and landmark figures will be kept separate. Estimates will be clearly labelled.

2. Dashboard Structure

The Power BI dashboard will have four sections. Each selected landmark will have its own page using a similar template.

Section	Purpose	Core Measures and Visuals
1. Tourism Overview	Track national tourism.	Arrivals; revenue; YoY growth; spending per tourist where available; tourist nights; monthly and annual trends.
2. Who Visits Egypt?	Analyze visitor source markets.	Arrivals by country/region; market share and growth; average stay; revenue per tourist where available; map and rankings.
3. Landmark Intelligence	Profile each selected landmark separately.	Visits and growth where available; location; tourism type; trends; seasonality; supported visitor and economic indicators.
4. Opportunities & Strategy	Turn findings into decisions.	Arrivals vs. revenue growth; seasonal gaps; potential markets; forecasts only when data supports them.

2.1 Selected Landmarks

Possible landmarks include the Pyramids of Giza, Grand Egyptian Museum, Karnak and Luxor Temples, Abu Simbel, Qaitbay Citadel, and Hurghada or Sharm El Sheikh. Final selection depends on reliable data.

2.2 Landmark Page Template

Component	Content
Landmark Profile	Name, image, location, governorate, and tourism type.
Performance Indicators	Available, validated visitor and growth figures.
Time Analysis	Monthly or annual trends, if available.
Seasonality	Observed high- and low-demand periods.
Visitor Profile	Source markets or visitor details, if available.
Economic Context	Relevant spending, accommodation, or revenue data.
Findings and Recommendations	Key insights and practical actions.

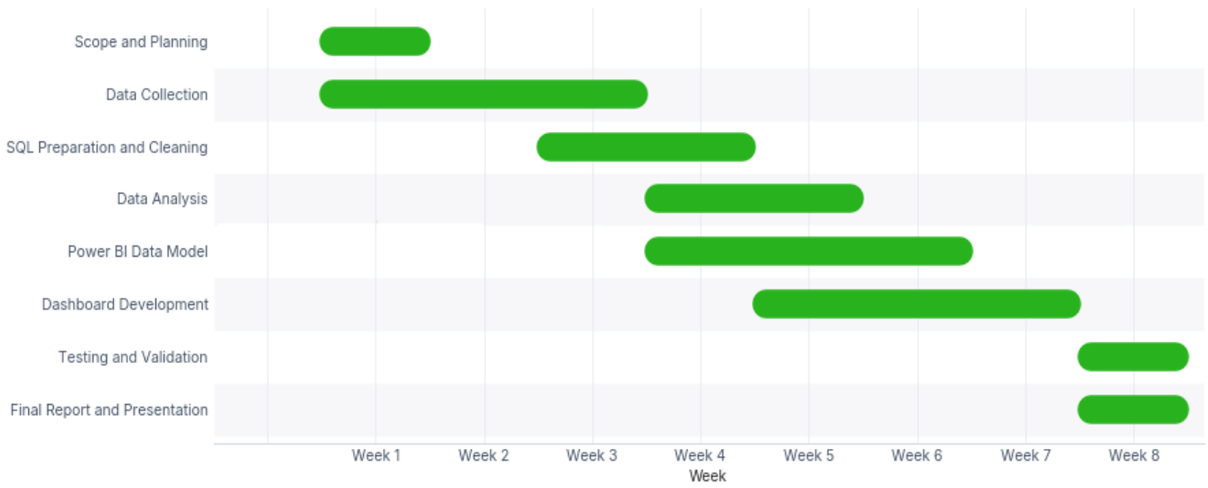
National or governorate figures will not be presented as landmark-level data

3. Project Plan and Schedule

The project is planned for eight weeks. Some activities may run in parallel.

Week(s)	Phase	Activities	Deliverable
1	Scope	Confirm objectives, scope, and roles.	Approved proposal
1–3	Data Collection	Collect and document reliable sources.	Source inventory and raw data
3-4	SQL and Cleaning	Create tables, standardize fields, remove duplicates, validate data.	Clean database
4-5	Analysis	Review trends, seasonality, and data gaps.	Initial findings
4–6	Data Model	Build Power BI relationships and measures.	Validated model and measures
5–7	Dashboard	Build the four sections and landmark pages.	Dashboard draft
8	Testing	Check measures, filters, navigation, and speed.	Tested dashboard
8	Finalization	Complete report and presentation.	Final report and presentation

Grannt Chart



3.1 Milestones

Milestone	Target
Scope approved	End of Week 1
Core data collected	End of Week 3
Database cleaned	End of Week 4
Initial analysis completed	End of Week 5
Model and measures validated	End of Week 6
Dashboard draft completed	End of Week 7
Final delivery	End of Week 8

4. Roles and Responsibilities

Role	Responsibilities	Output
Project Manager / Research Lead	Manage scope, schedule, sources, and risks.	Plan and source inventory
SQL / Data Engineer	Build tables, clean data, and check quality.	SQL database
Data Analyst	Analyze trends, seasonality, and limitations.	Findings and report
Power BI Developer	Build data model, measures, and overview/markets/strategy pages.	Interactive dashboard
Landmark Analyst / BI Designer	Research landmarks and build their pages.	Landmark pages and recommendations

All members will help validate data, test the dashboard, and prepare the final presentation.

4.1 Resources

Resources: SQL Server, Power BI Desktop, Excel/CSV files, reliable tourism sources, and shared project storage.

5. Risk Assessment and Mitigation

Risk	Impact	Mitigation
No landmark-level visit data	High	Select landmarks with reliable data; label regional figures.
No landmark revenue data	High	Report revenue only at its supported level.
Different source periods or definitions	High	Document sources and compare like with like.
Duplicates or inconsistent names	Medium	Standardize names and run quality checks.
Scope grows too large	High	Start with a fixed landmark list.
Incorrect relationships or measures	High	Document and validate the data model.
Team delays	Medium	Assign owners and review progress weekly.
Estimates appear official	High	Label estimates and record assumptions.

6. Suggested Key Performance Indicators

6.1 Tourism KPIs

KPI	Definition / Application
Total Tourist Arrivals	Reported arrivals for the selected period.
Tourism Revenue	Revenue reported by the source for its stated period.
YoY Arrivals Growth %	(Current arrivals − previous arrivals) ÷ previous arrivals × 100.
Average Spending per Tourist	Revenue ÷ compatible tourist count.
Total Tourist Nights	Reported tourist nights for the selected period.
Average Length of Stay	Tourist nights ÷ compatible visitor count.
Market Share %	Market arrivals ÷ total arrivals × 100.
Seasonality Index	Monthly demand compared with a documented baseline.
Landmark Visits Growth %	Growth in measured landmark visits, if a reliable time series exists.
Forecast Accuracy	Compare forecast values with actual values on a test period.

6.2 Project KPIs

KPI	Proposed Target	Measurement
Data Quality Rate	At least 95%	Records passing validation checks.
Dashboard Completion	100%	All agreed sections and landmark pages delivered.
KPI Validation Rate	100%	Published measures checked against sources.
Dashboard Response Time	5 seconds or less	Core interactions on the test setup.
On-Time Delivery	100%	Milestones completed on schedule.

These are proposed targets, not achieved results.

7. Final Deliverables and Acceptance Criteria

7.1 Deliverables

- Project Proposal and Project Plan, including timeline, responsibilities, and risks.
- Data Source Inventory documenting sources, fields, reporting periods, and limitations.
- SQL database, table definitions, and data-preparation queries.
- Power BI data model and documented KPI measures.
- Interactive Power BI dashboard with four main sections and dedicated pages for the selected landmarks.
- Analytical report summarizing findings, limitations, and recommendations.
- Final presentation for project review.

7.2 Acceptance Criteria

- All sources and material assumptions are documented.
- Database tables, relationships, and core calculations have been validated.
- Dashboard filters and navigation operate as intended.
- Landmark pages use data at the correct geographic and statistical level.
- Missing data, estimates, and analytical limitations are clearly disclosed.
- Recommendations are traceable to observed findings.

8. Conclusion

Egypt Tourism Intelligence combines national tourism analysis, source-market insights, dedicated landmark pages, and practical recommendations. The project uses SQL Server and Power BI over eight weeks. Final results depend on reliable, comparable data.